However, we humans also have a history of making long-lasting social change that nearly everyone thought impossible. Since our earliest days, we've worked not just for our own advantage but for our children and our grandchildren. My grandparents and parents did so for me. We have often, despite our diversity, despite endemic hatreds, pulled together to face a common enemy. We seem, these days, much more willing to recognize the dangers before us than we were even a decade ago. The newly recognized dangers threaten all of us equally. No one can say how it will turn out down here.

THE MOON WAS WHERE the tree of immortality grew in ancient Chinese myth. The tree of longevity if not of immortality, it seems, indeed grows on other worlds. If we were up there among the planets, if there were self-sufficient human communities on many worlds, our species would be insulated from catastrophe. The depletion of the ultraviolet-absorbing shield on one world would, if anything, be a warning to take special care of the shield on another. A cataclysmic impact on one world would likely leave all the others untouched. The more of us beyond the Earth, the greater the diversity of worlds we inhabit, the more varied the planetary engineering, the greater the range of societal standards and values—then the safer the human species will be.

If you grow up living underground in a world with a hundredth of an Earth gravity and black skies through the portals, you have a very different set of perceptions, interests, prejudices, and predispositions than someone who lives on the surface of the home planet. Likewise if you live on the surface of Mars in the throes of terraforming, or Venus, or Titan. This strategy—breaking up into many smaller self-propagating groups, each with somewhat different strengths and concerns, but all marked by local pride—has been widely employed in the evolution of life on Earth, and by our own ancestors in particular. It may, in fact, be key to understanding why we humans are the way we are.* This is the second of the missing justifications for a permanent human presence in space: to improve our chances of surviving, not just the catastrophes we can foresee,